

Portfolio Risk Measurement and VaR Backtesting in R: Does Diversification Protect Investors During Market Stress?

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GitHub Repository:

<https://github.com/weitaolim986-jpg/portfolio-risk-analysis>

Executive Summary

This report aims to examine the effects of portfolio diversification on portfolio risk, and explore multiple models in producing risk estimates to understand whether the complexity of a model affects its performance in estimating risk. The assets used are SPY, IEF, and GLD, as the proxies of the U.S. equities, intermediate-term U.S. Treasury bonds, and gold to construct Equity-Heavy, Balanced, and Defensive portfolios. From the analysis of data from 2015 to 2025, greater allocation to bonds and gold reduced volatility, maximum drawdown, VaR, and Expected Shortfall, but at the cost of lower long-term returns. Stress testing also shows that diversification reduced losses and drawdown, but the degree of protection depended on market context. The results of rolling backtesting and comparison across Historical VaR, Parametric VaR, Normal GARCH VaR, and Student-t GARCH VaR also suggest that more complex VaR models do not uniformly perform better at calibrating VaR forecasts. However, the analysis could be extended by using a longer historical sample, more asset classes, alternative portfolio weights, formal VaR calibration tests and broader backtesting settings.

1. Introduction and Motivation

1.1 Background

Investors care about both return and risk of their investment portfolios. While a portfolio with a high long-term return may seem appealing, it is important to acknowledge that it might suffer severe losses during stressed periods or market downturns. Therefore, investors diversify their portfolios to keep investing in the high-return assets while counteracting the potential losses with other assets that do not move in the same direction. Beyond comparing the returns and risks of portfolios, risk management in finance also requires tools for estimating downside losses, letting the investors be prepared for the “worst case scenario”.

1.2 Motivation

As a data science student passionate about applied mathematics and statistics, I am keen to explore the field of quantitative finance and risk management, essential tools in driving the development and innovation of the vibrant financial sector. To deepen my exposure to the area, I chose to study portfolio risk over predicting stock prices because risk analysis provides an interpretable framework to examine diversification, downside losses, and model reliability in estimating portfolio risk. This is relevant to risk management, as the objective is not only to pursue high returns, but also to quantify exposure to adverse market movements. The use of equities, Treasury bonds and gold in this report is supported by the low correlations among them, reducing the effect on portfolio values under extreme conditions causing the value of one of the instruments to drop. Therefore, they form a meaningful introductory diversification study.

1.3 Why VaR modelling and backtesting matter

Value at Risk (VaR) is an estimated loss threshold for an investment over a specified period at a chosen confidence level (CQF, n.d.). For example, a one-day 95% VaR corresponds to a loss threshold exceeded on approximately the worst 5% of observed trading days, while daily losses remain below the threshold on the remaining 95% of trading days. However, different VaR models may generate different estimates of the losses and risks, necessitating the evaluation of each risk model against realized losses instead of blanket acceptance. This motivates the use and comparison of Historical VaR, Parametric VaR, Normal GARCH (Full: Generalized Autoregressive Conditional Heteroskedasticity) VaR, and Student-t GARCH VaR on their estimates based on the dataset in this report. GARCH is a model that computes conditional variance based on past squared observations and past conditional variances,

allowing the “volatility clustering” phenomenon (Marisetey, 2024). It was implemented in R through the `rugarch` package (Galanos, 2026).

1.4 Research Questions

1. Does adding Treasury bond and gold exposure to an equity-heavy portfolio reduce volatility, drawdowns and tail losses?
2. How do different portfolio allocations affect the trade-off between long-term return and downside protection?
3. How do Historical VaR, Parametric VaR, Normal GARCH VaR and Student-t GARCH VaR differ in their estimates of downside risk?
4. How reliably do these VaR models forecast one-day-ahead portfolio losses in rolling backtesting?
5. Does diversification provide similar protection across different historical stress periods?

2. Data and Portfolio Construction

2.1 Data Selection

This report uses the historical price data of SPY, IEF, and GLD from 2015 to 2025 from Yahoo Finance (Yahoo Finance, n.d.) accessed through the `quantmod` package in R (Ryan & Ulrich, 2025). Daily observations and fluctuations are studied to fully investigate the micro-level movements in prices. One important setup is that adjusted closing prices are used because they account for relevant corporate actions and distributions, such as stock splits and dividends, thereby providing a more suitable basis for calculating historical investment returns. Moreover, SPY, IEF, and GLD are purposefully selected for their different roles in a portfolio, as shown in the following table:

Asset	ETF	Role in Portfolio
US equities	SPY	Growth-oriented equity exposure
US Treasury bonds	IEF	Lower-risk fixed-income exposure
Gold	GLD	Alternative defensive/diversifying exposure

2.2 Portfolio Design

This report explores three portfolios, containing the same three assets but varying by the proportion of equities versus defensive assets. The changing weights of the three assets change how conservative and diversified the portfolio is, allowing us to examine the effects of diversification on its return and risk. The following table shows exactly the three portfolios used and the intended risk profile they represent.

Portfolio	SPY	IEF	GLD	Intended Risk Profile
Equity-Heavy	80%	10%	10%	Higher growth, higher risk
Balanced	50%	25%	25%	Intermediate risk-return profile
Defensive	30%	40%	30%	Lower equity exposure, greater defensive allocation

2.3 Portfolio Assumptions

All three portfolios are represented using fixed target weights, implying the need for regular rebalancing to maintain the asset allocations. Furthermore, transaction costs, taxes and Exchange-Traded Funds (ETF) fees are not included, which might shrink the actual return with regular rebalancing. Returns are also evaluated in USD, not accounting for the effects of exchange rate movements on the returns in SGD.

3. Methodology

3.1 Daily Returns and Normalized Asset Growth

The following equation is the formula used to calculate daily returns:

$$r_t = \frac{P_t}{P_{t-1}} - 1$$

where r_t denotes the daily return on day t, P_t denotes the adjusted closing price on day t, and P_{t-1} denotes the adjusted closing price on the previous trading day. It measures percentage change rather than dollar price levels, making assets comparable. Normalized asset values show how an initial investment of \$100 would have evolved over time if invested in the asset. However, the normalized growth is only used for visual comparison, while daily returns are used for risk calculations.

3.2 Portfolio Return Calculation

Portfolio daily returns are calculated as the weighted combination of asset daily returns:

$$r_{p,t} = \omega_1 r_{1,t} + \omega_2 r_{2,t} + \omega_3 r_{3,t}$$

where $r_{p,t}$ denotes the portfolio return on day t, ω_i denotes the weight assigned to each asset, and $r_{i,t}$ denotes the daily return of each asset. In R, the calculation is implemented through matrix multiplication.

3.3 Return and Ordinary Risk Measures

Compounded Annualized Return

This represents the constant yearly growth rate equivalent to the portfolio's actual compounded growth over the study period. To obtain the compounded annualized return, each daily return is first converted into a growth factor, $1 + r_t$. Then, all growth factors over the period are multiplied to obtain the cumulative growth multiplier. Finally, the multiplier is annualized by raising it to the reciprocal of the number of years analyzed, calculated by dividing the number of trading days throughout the selected data by the number of trading days in a year, i.e. 252, then subtracting 1 from the annualized result. This formula summarizes this process using mathematical notions:

$$\text{Annualized Return} = \left(\prod_{t=1}^T (1 + r_t) \right)^{1/(T/252)} - 1$$

Where $\prod_{a=1}^b x_a$ represents the product of $x_1, x_2, \dots, x_b$.

Annualized Volatility

It measures the typical fluctuation in daily returns, scaled to an annual basis. It is calculated as the product of the square root of the number of trading days in a year, usually $\sqrt{252}$, and daily volatility, σ .

Maximum drawdown

It measures the largest fall in portfolio value from a previous peak to a subsequent trough. Based on daily returns, a cumulative wealth path is constructed to determine running historical peaks, which are used to calculate percentage decline from peak. The worst decline is selected for each portfolio to reflect the most severe historical reduction in portfolio wealth to investors.

3.4 Correlation and Diversification Analysis

Correlation measures the extent to which asset returns move together. Thus, this report uses the assets with low or negative correlation to benefit from portfolio diversification. However, correlation does not prove causation or guarantee future protection to an investment.

3.5 Value at Risk and Expected Shortfall

Historical VaR is the historical loss threshold exceeded only in the worst tail of observed days based on historical data. It is computed from the specified empirical lower-tail quantile of portfolio returns, expressed as a positive dollar loss.

Expected Shortfall is the average loss on days when the loss exceeds the VaR threshold (Privault, 2026). It complements VaR, indicating how severe losses were beyond the VaR threshold. A crucial assumption is that all dollar risk estimates are based on a hypothetical USD 10,000 portfolio value.

3.6 Historical VaR versus Parametric VaR

This report involves the comparison of the two stated methods to evaluate VaR. It aims to assess whether VaR estimates depend on method choice, examine whether focusing on more extreme losses increases differences between models, and motivates the later use of dynamic-volatility and heavy-tail models. They differ in their core ideas and assumptions, as shown in the following table:

Method	Core Idea	Main Assumption
Historical VaR	Uses actual past return distribution	Future-like tail behaviour is represented by observed historical outcomes
Parametric VaR	Uses estimated mean and standard deviation	Returns are modelled using a normal-distribution assumption

Additionally, there also exists a formula to compute the Parametric VaR:

$$VaR_c = -(\mu + z_{1-c}\sigma)V_0$$

Where μ denotes the mean, σ denotes the standard deviation, z denotes the Z-statistic of the difference between 1 and the given confidence level. V_0 is the assumed portfolio value, e.g. \$10,000 for a portfolio.

A histogram and Q-Q plot are also used as an exploratory check of whether portfolio returns resemble a normal distribution.

3.7 Rolling VaR Backtesting

Backtesting is used to check whether realized losses exceed its predicted VaR threshold more often than expected. The method uses an initial historical window of 500 trading days to estimate one-day-ahead VaR for the following trading day. The realized next-day portfolio return is then observed and an exceedance will be recorded if realized loss is greater than predicted VaR. This process is repeated through the testing period. The expected exceedance rates are derived directly from the difference between 100% and the confidence levels.

If the calculated rate is substantially above expected, the model may underestimate downside risk, while it may be too conservative if the rate is substantially below expected. Following this logic, if the rate is close to expected, the model appears reasonably calibrated within the sample.

3.8 GARCH Conditional Volatility Modelling

This model uses the concept of volatility clustering, meaning that volatile market periods may be followed by further volatility. At times when the assets are volatile over a specific period, a fixed or purely sample-based risk estimate may not react to changing conditions in the same way as a dynamic volatility model. However, GARCH estimates one-day-ahead volatility based on current volatility and other variables, as shown in the following standard GARCH(1,1) variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

where each variable is explained by the following table:

Term	Meaning
σ_t^2	Estimated conditional variance for day t
ω	Baseline variance component
ε_{t-1}^2	Most recent squared unexpected return shock

σ_{t-1}^2	Previous estimated conditional variance for day t -1
α_1	Sensitivity to new shocks
β_1	Persistence of volatility

However, this report uses two different GARCH models, whose differences shall be made clear to understand the reason behind the different results from them. Normal GARCH VaR assumes the standardized shocks follow the normal distribution, whereas Student-t GARCH assumes the shock to follow Student-t distribution to allow heavier tails. In the analysis of this report, a rolling window of 500 trading days and a refitting frequency of 1 day are used, i.e. the parameters in the formula are re-estimated before each one-day-ahead forecast using the updated rolling estimation window. In spite of this, they both use one-day ahead conditional mean and conditional volatility forecasts from the model, with only difference in distributional assumptions.

3.9 Historical Stress Testing

To assess the protection of each portfolio, two periods of serious market fluctuations are selected for stress testing. The following table explains the purpose behind the choice of each stress period.

Stress Period	Time Window Used	Purpose
COVID-19 sell-off	19/2/2020 - 23/3/2020	Examine portfolio behaviour during a sharp equity-led market collapse
2022 inflation / rising-rate environment	1/1/2022 - 31/12/2022	Examine portfolio behaviour when equities and bonds faced pressure in a different macroeconomic environment

During each period, the cumulative return and maximum drawdown for each portfolio are calculated and used to compare with the performance of the portfolios during normal periods.

4. Results and Analysis

4.1 Individual Asset Performance and Correlation

To evaluate the performance of each asset over the specified time period, an asset summary table showing the return and volatility of each of the assets as follows is used. It is complemented by a graph showing the growth of \$100 for all three assets.

Asset	Mean daily return (%)	Daily volatility (%)	Annualized return (%)	Annualized volatility (%)
SPY	0.06	1.12	14.29	17.79
IEF	0.01	0.42	1.49	6.63
GLD	0.05	0.93	12.50	14.75

Table 1: Asset Summary

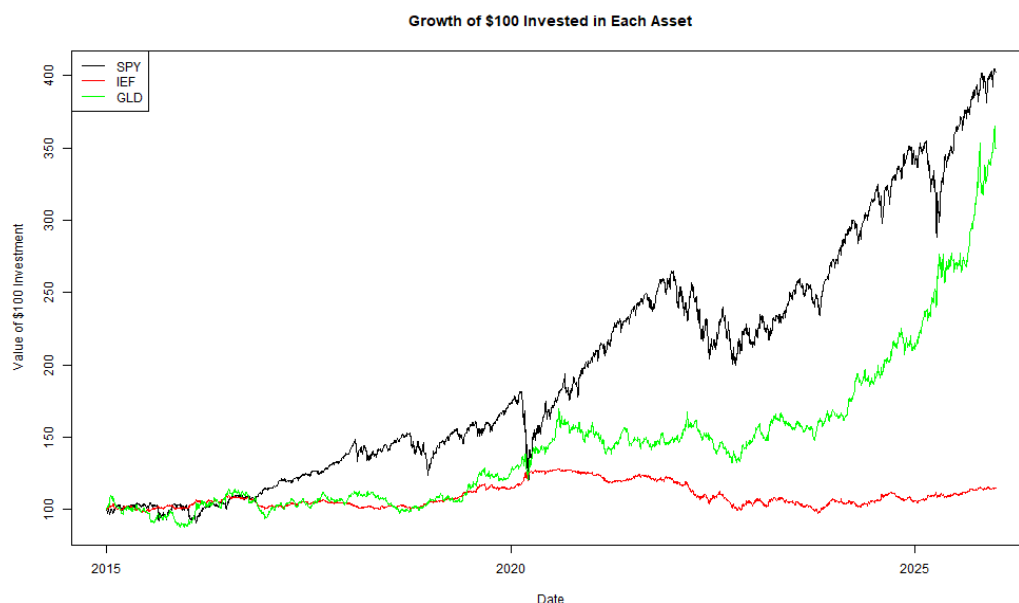


Figure 1: Growth of \$100 for SPY, IEF, and GLD

From the graph and table, SPY yields the highest return, but at the cost of higher return volatility and price uncertainty. GLD's return and volatility are in between SPY and IEF, while IEF is the least lucrative but most stable in price. Besides return and volatility, the correlation patterns among the assets are also observed and summarized in the following heatmap:

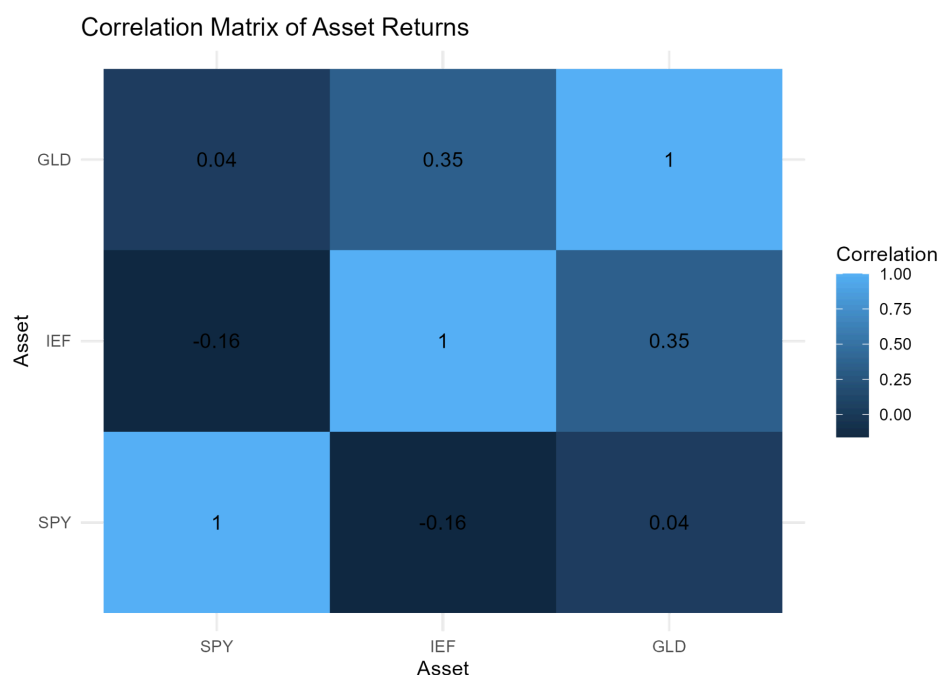


Figure 2: Correlation Heatmap of SPY, IEF, and GLD

According to the heatmap, there does not exist a strong correlation between any two of the assets, indicating that portfolios containing these assets may bring more diversification benefits.

4.2 Portfolio Risk-Return Comparison

The three assets are combined to construct three different portfolios, namely equity-heavy, balanced, and defensive portfolios. The risk and return of each portfolio are shown in the following table of the annualized return, volatility, and maximum drawdown of each portfolio. It is complemented by a graph depicting the growth of \$10,000 invested in each portfolio.

Portfolio	Annualized return (%)	Annualized volatility (%)	Maximum drawdown (%)
Equity heavy	12.53	14.31	-27.42
Balanced	10.69	9.90	-18.45
Defensive	8.69	7.81	-16.91

Table 2: Portfolio Summary

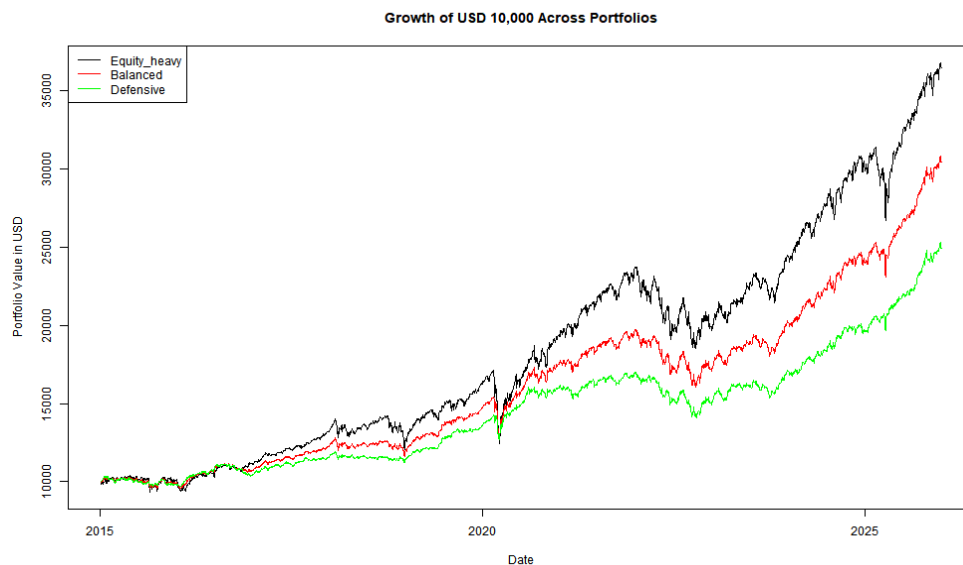


Figure 3: Growth of \$10,000 for three portfolios with varying emphases on equities

According to the graph and table, the equity-heavy portfolio generates the highest return and most significant growth in value, but also carries the highest volatility and maximum drawdown, implying larger losses for investors. While the return of the defensive portfolio is much lower than that of the equity-heavy one, it brings the lowest volatility and maximum drawdown, thereby reducing the risks investors face. Thus, greater allocation of defensive assets helped reduce risks and losses, although accompanied by lower returns.

4.3 Historical VaR and Expected Shortfall

To estimate the risks and potential losses of each portfolio, VaR and Expected Shortfall are calculated and summarized in the following table and graph:

Portfolio	VaR at 95% (\$)	Expected Shortfall at 95% (\$)
Equity-heavy	131.15	215.22
Balanced	91.68	144.66
Defensive	73.66	111.57

Table 3: VaR and Expected Shortfall table



Figure 4: VaR and Expected Shortfall bar chart for each portfolio

From the above table and graph, it is obvious that the equity-heavy portfolio carries the highest downside risk, followed by the balanced and defensive portfolio. Expected Shortfall is larger than VaR in this analysis because it measures the average loss conditional on losses exceeding the VaR threshold, rather than reporting only the threshold itself. This result shows that portfolio diversification does reduce estimated tail losses, as the defensive portfolio records the lowest in both VaR and Expected Shortfall.

4.4 Historical Stress Testing

To thoroughly assess the performance of each portfolio, a stress testing is conducted by selecting two stress periods, namely the COVID-19 sell-off and hike in interest rate in 2022. The results are shown in the following table:

Portfolio	COVID-19 Return	COVID-19 Maximum Drawdown	2022 Return	2022 Maximum Drawdown
Equity-Heavy	-27.10%	-27.42%	-15.84%	-21.78%
Balanced	-16.97%	-17.38%	-12.61%	-18.39%
Defensive	-9.34%	-11.08%	-11.35%	-16.91%

Table 4: Results of stress testing on the three portfolios

From the table, diversification reduces losses by a significant 65% and drawdown by 60%, showing a strong diversification benefit even during COVID-19. However, in 2022, the reduction effect drops to 28% and 22% respectively. This pattern is consistent with the role of Treasury bonds as defensive assets during the COVID-19 equity sell-off, as IEF remained relatively resilient while SPY declined sharply (Shu et al., 2025). This partially recovered portfolio value from the stock crash. The following graph shows the price movements of the three assets during the COVID-19 stock market crash.

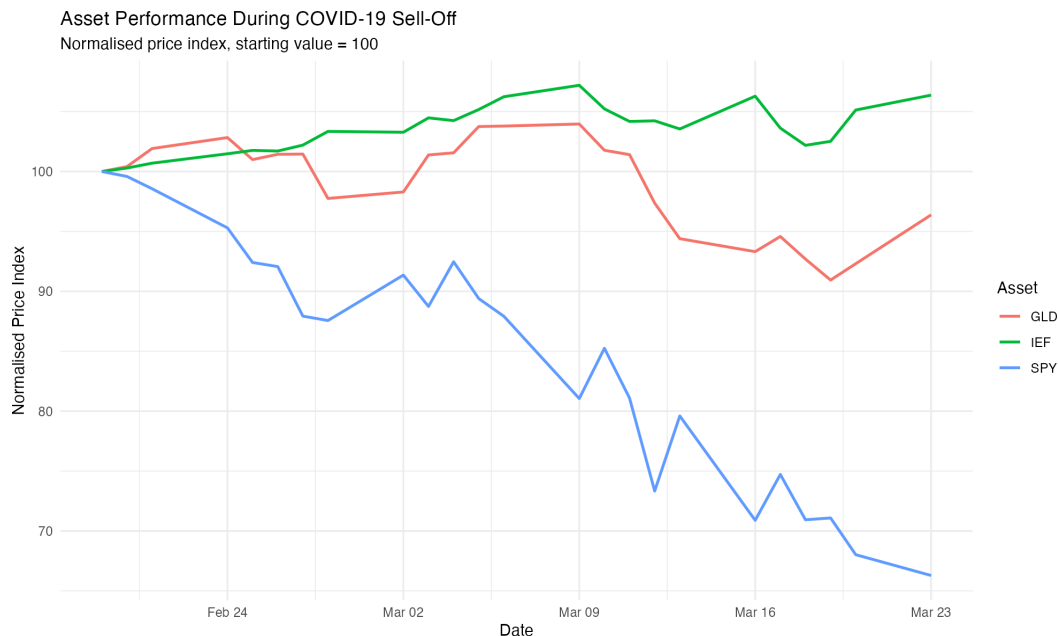


Figure 5: Normalized price movements of SPY, IEF, and GLD during the COVID-19 equity sell-off

From the graph, it is noticeable that SPY depreciates substantially from 100 to lower than 70, while IEF remains strong and appreciates overall throughout the period. Therefore, the defensive portfolio consisting $\frac{1}{3}$ of bonds suffered lower losses. However, amid the interest rate rise in 2022, IEF also depreciated, weakening the protection expected from a higher bond allocation, hence the defensive portfolio did not provide protection as strong as before. This can be illustrated by the following graph depicting the movements of normalized prices for all three assets.

Asset Performance During 2022 Inflation and Rising-Rate Environment
Normalised price index, starting value = 100



Figure 6: Normalized price movements of SPY, IEF, and GLD during the 2022 rising-interest-rate environment

As the graph shows, both SPY and IEF declined in value, weakening the protection against portfolio value loss gained through diversification. Therefore, the context of stress periods matters when evaluating defensive allocations as the assets perceived as defensive might be affected seriously under certain circumstances, thus affecting the performance of such portfolios.

4.5 Historical VaR versus Parametric VaR

This report uses different models to compute the VaR of each portfolio, results of which are shown in the following graph:

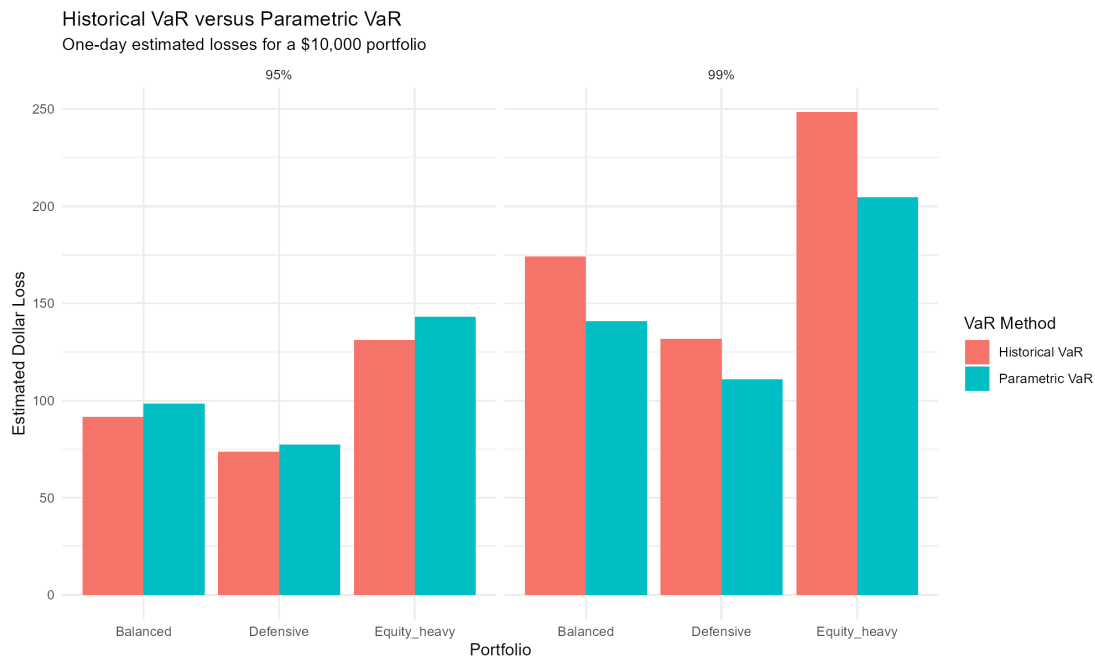


Figure 7: Comparison between historical and Parametric VaR models for a \$10,000 portfolio of three different types

These two models invariably rank the portfolios the same way in terms of their VaRs at both 95% and 99% confidence levels, i.e. the equity-heavy portfolio still appears to be the riskiest among the three.

However, the parametric model results in higher VaRs at 95% confidence level for all three portfolios, while the historical model generates higher estimates at 99% confidence level. What's more worth noticing is that the difference between the estimates from the two models is bigger at 99% than at 95% confidence level. The Equity-Heavy portfolio provides an illustrative example of why the Historical and Parametric VaR estimates may diverge, particularly at more extreme confidence levels.

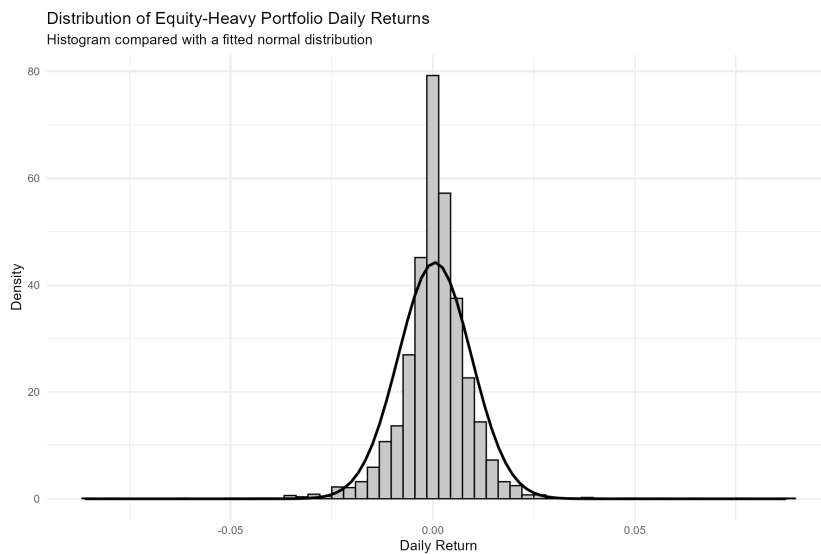


Figure 8: Comparison between normal distribution and actual distribution of equity-heavy portfolio's daily returns

Taking the equity-heavy portfolio as an example, the distribution appears more heavily concentrated around returns close to zero, while also containing extreme observations in the tails not well captured by the fitted normal curve. To assess the deviation of the return from normal distribution more rigorously, a normal Q-Q plot is constructed, suggesting deviations from normality in both tails, providing exploratory evidence that extreme return observations are not fully captured by the fitted normal distribution. Therefore, when VaR is reported as a positive dollar loss, Parametric VaR is slightly larger at 95%, while Historical VaR is substantially larger at 99%, justifying the differences observed in the bar chart above. The differences in assumptions and tail risk estimates between these two models motivate the use and comparison of GARCH-based VaR models, which allow conditional volatility to change over time. Their forecasting performance is then evaluated through rolling backtesting.

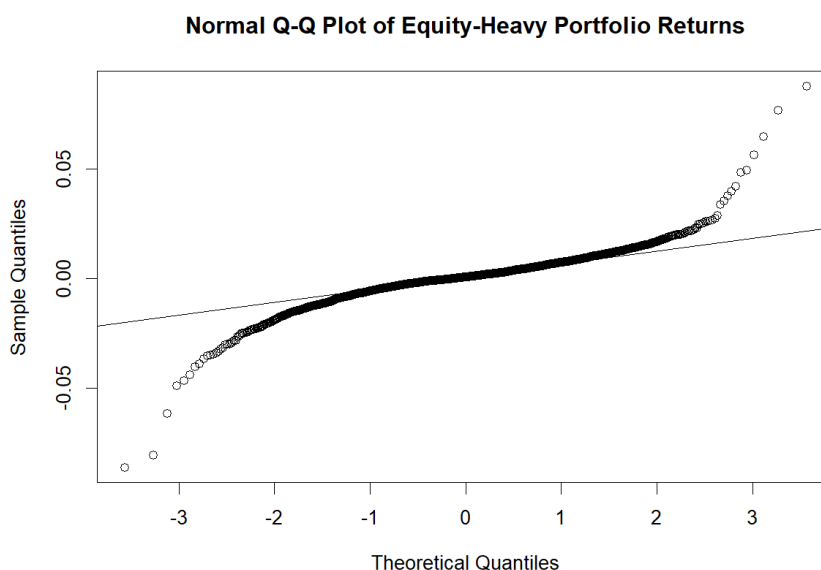


Figure 9: Normal Q-Q plot of the equity-heavy portfolio's returns

4.6 GARCH Conditional Volatility Results

Normal GARCH is used to estimate the conditional volatility in each portfolio's return. The following graph depicts the conditional volatility for the equity-heavy portfolio.

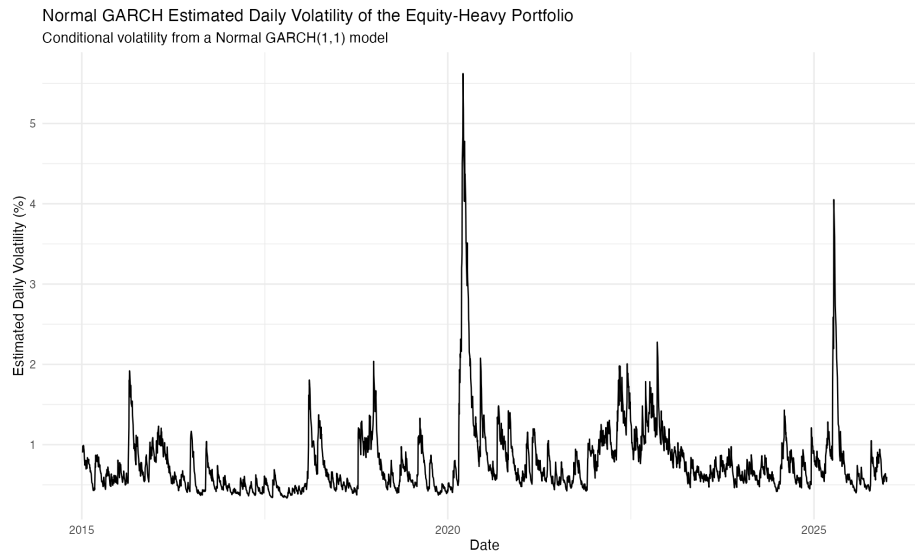


Figure 10: Normal GARCH estimated volatility of the equity-heavy portfolio

According to the graph, volatility changes materially over time with some major fluctuations. Major spikes expectedly occurred during stress periods such as the COVID-19 equity sell-off, when daily volatility was estimated to be nearly 6%. For the portfolio data used in this report, it is reasonable to model time-varying volatility as asset prices, and thus risks are largely affected by recent economic and financial uncertainties.

4.7 Four-Model VaR Backtesting Comparison

To assess whether complex models produce better-calibrated one-day-ahead VaR forecasts, this report further compares the observed exceedance rates produced by each model for each portfolio with their expected benchmarks. The following graph and numerical table illustrate the comparison:

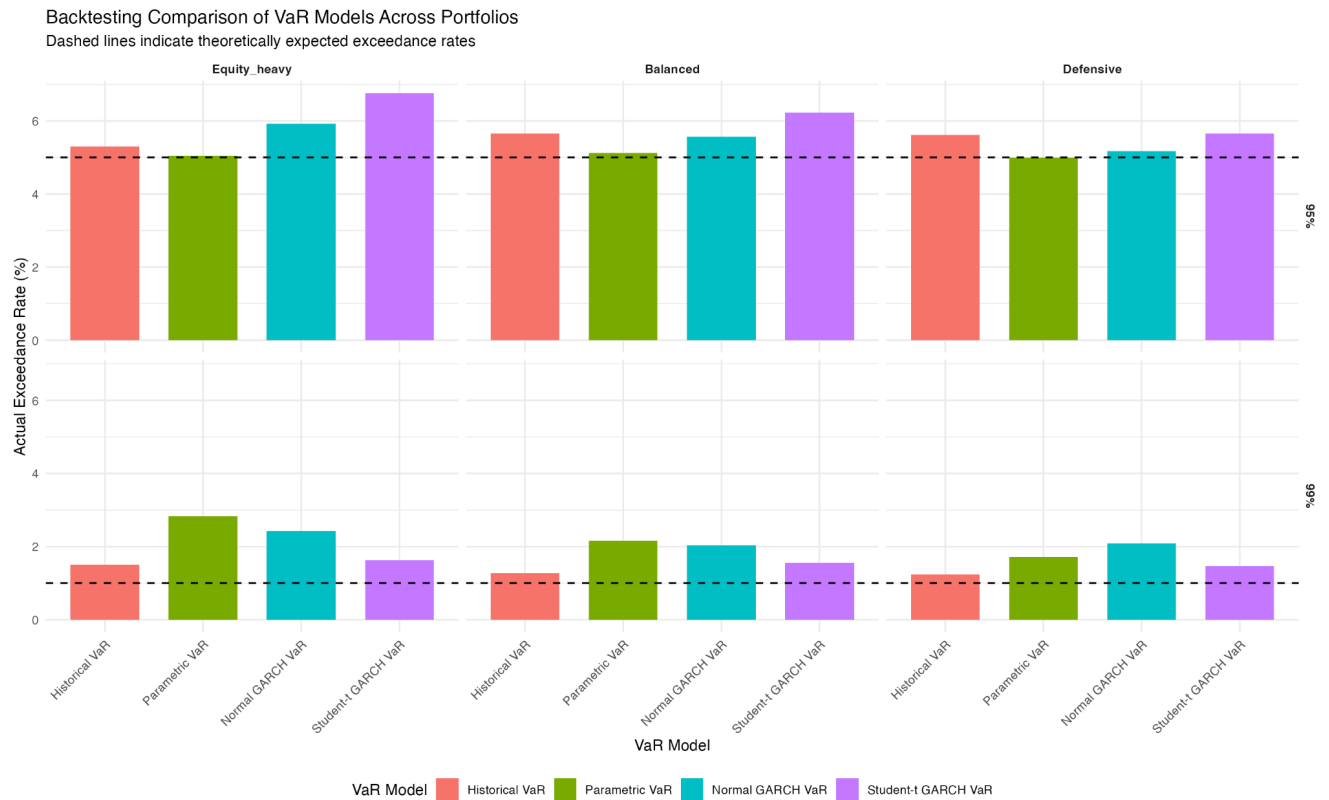


Figure 11: Backtesting comparison of all four VaR models across portfolios and both confidence levels. Dashed line indicates expected exceedance rates.

Portfolio	Confidence Level	Method	Number of Exceedances	Number of Test Days	Exceedance Rate Percent	Expected Exceedance Rate Percent
Balanced	95%	Normal GARCH VaR	126	2264	5.57	5
Balanced	95%	Historical VaR	128	2264	5.65	5
Balanced	95%	Parametric VaR	116	2264	5.12	5
Balanced	95%	Student-t GARCH VaR	141	2264	6.23	5
Balanced	99%	Normal GARCH VaR	46	2264	2.03	1
Balanced	99%	Historical VaR	29	2264	1.28	1

Balanced	99%	Parametric VaR	49	2264	2.16	1
Balanced	99%	Student-t GARCH VaR	35	2264	1.55	1
Defensive	95%	Normal GARCH VaR	117	2264	5.17	5
Defensive	95%	Historical VaR	127	2264	5.61	5
Defensive	95%	Parametric VaR	113	2264	4.99	5
Defensive	95%	Student-t GARCH VaR	128	2264	5.65	5
Defensive	99%	Normal GARCH VaR	47	2264	2.08	1
Defensive	99%	Historical VaR	28	2264	1.24	1
Defensive	99%	Parametric VaR	39	2264	1.72	1
Defensive	99%	Student-t GARCH VaR	33	2264	1.46	1
Equity heavy	95%	Normal GARCH VaR	134	2264	5.92	5
Equity heavy	95%	Historical VaR	120	2264	5.3	5
Equity heavy	95%	Parametric VaR	114	2264	5.04	5
Equity heavy	95%	Student-t GARCH VaR	153	2264	6.76	5

Equity heavy	99%	Normal GARCH VaR	55	2264	2.43	1
Equity heavy	99%	Historical VaR	34	2264	1.5	1
Equity heavy	99%	Parametric VaR	64	2264	2.83	1
Equity heavy	99%	Student-t GARCH VaR	37	2264	1.63	1

Table 5: Numerical table of the comparison among all four models for all portfolios and both confidence levels

Between Historical and Parametric VaR, the former was less closely calibrated at the 95% confidence level than the latter, while becoming more closely calibrated at the 99% confidence level. Both Parametric VaR and Normal GARCH VaR became less closely calibrated at 99% than at 95%, but the former produces exceedance rates very close to the expected 5% benchmark. When heavier tails are incorporated, the calibration performance shows a strong contrast: it improves and ranks second across all portfolios at 99%, while being almost the worst at 95%.

Comparing across portfolios, the relative performance of the models is broadly similar, but the ranking changed noticeably between the 95% and 99% settings. Furthermore, a pattern worth noticing across both confidence levels is the difficulty in modelling the equity-heavy and thus riskiest portfolio. From the graph, the largest deviations from expected exceedance rates at both confidence levels come from the equity-heavy portfolio. Perhaps more unexpectedly, models that underestimate risks include the sophisticated models such as the Student-t GARCH VaR, meaning that more sophisticated models do not uniformly perform better at forecasting downside risks.

5. Discussion

5.1 Did Diversification Reduce Risk?

From previous analysis on historical data, the most defensive and diversified portfolio with the lowest equity allocation and greatest combined bond-and-gold allocation reports the lowest ordinary volatility, maximum drawdown, VaR and Expected Shortfall. It also survives the stress testing, showing a better performance in all the four metrics under periods of extreme market stress. Therefore, diversification reduces historical downside risk, but whether it guarantees protection in every crisis still remains questionable.

5.2 Did Protection Depend on the Market Environment?

Despite a better performance under stress periods, the magnitude of improvement for the defensive and diversified portfolio drops during 2022's rising-rate environment compared to the COVID-19 sell-off. This is caused by the negative effects of the rising interest rate environment on the price/value of the defensive asset in the portfolio, or in this report, bonds, thus reducing the value of the defensive portfolio which puts a larger emphasis on the defensive asset. Therefore, a defensive allocation of assets may protect strongly during one type of shock, but less strongly in another.

5.3 Did More Complex Risk Models Perform Better?

The performance of the four models at forecasting downside risk suggests that a dynamic way of computing conditional volatility does not necessarily produce better estimates. However, heavy-tailed modelling does help under more extreme observations, as shown by its improved performance at 99% for all three portfolios. This analysis also prompts us to rethink the power of the simple and seemingly least robust Historical VaR, as it appears to produce the most consistent and stable results across portfolios and confidence levels. Thus, the modelling performance of a model should be judged through backtesting, rather than technical complexity alone.

6. Limitations and Future Work

6.1 Data and Asset Scope

This report uses historical data, which is not representative of what happened during other historical windows and in the future. The type and severity of events happening in the future might not resemble what happened from 2015 to 2025, resulting in different analytical outcomes.

Furthermore, SPY, IEF, and GLD are all simplified proxies of assets, which is not fully representative of the nature and movements of the asset each of them belongs to. Thus, the performance recorded in each asset might differ from what would have happened to another proxy of the same asset.

All the metrics calculated are denominated in USD, thus excluding the effects of USD-SGD exchange rate movements on the returns, losses, and other measures denominated in SGD. Therefore, the results for the Singaporean market might change even if market conditions remain unchanged.

6.2 Portfolio Assumptions

Portfolio weights are assumed to be fixed, requiring regular rebalancing by investors. Otherwise, the returns and risks might deviate from the results if portfolio weights change. Furthermore, transaction costs, taxes and management fees are not included in the analysis, inclusion of which might alter the results significantly especially if portfolios are regularly rebalanced.

6.3 Risk-Model Limitations

There are also certain limitations to the VaR models used in the report. The Historical VaR depends on observed historical losses, thus not capable of capturing every future crisis or how portfolios might grow after the observed periods. Parametric VaR assumes returns follow a normal distribution for its analysis of volatility, but this assumption is not always true, and when false, can cause results to deviate even more from the actual observations. On the other hand, standard GARCH treats positive and negative shocks symmetrically, which is not necessarily true as negative shocks can affect portfolios' value even more than positive ones do. Although heavy-tailed models may be considered, it is crucial to remember that Student-t GARCH is only one possible model of this kind, and that there might be better, more robust heavy-tailed models.

6.4 Backtesting and Stress Testing Scope

Backtesting and stress testing use different lengths of rolling window, which might capture different happenings and each window is not representative of what had happened before it and what would happen next. The 95% and 99% confidence levels are chosen because they are commonly used in financial risk analysis. However, performances might vary if confidence levels change from the typical thresholds.

Moreover, exceedance rates are used extensively in this report to measure calibration, but might not fully represent the calibration of the models as exceedance rates change by other factors. The selected stress periods for stress testing do not represent the resilience of portfolios in all possible future crises, hence the results are also not applicable to all future events.

Future Work

To improve the reliability and accuracy of this report, some improvements are given as a reference for future report writing. Firstly, the return and risk analysis should be denominated in SGD, thus accounting for exchange rate movements to allow readers to directly understand how lucrative or volatile portfolios in Singapore's market are. Transaction and other miscellaneous fees should also be adjusted, especially if portfolios are regularly rebalanced to ensure the returns shown are more accountable and reliable. Furthermore, VaR calibration tests should also be done formally, such as through Kupiec testing. To address the symmetric volatility assumed by the standard GARCH (1,1) model, asymmetric volatility models such as GJR-GARCH or EGARCH might also be used to observe how volatility changes as the shock assumptions of a model change. To dive deeper into portfolio construction and resilience, additional assets shall be included in each portfolio and more diverse stress scenarios shall be applied to examine the actual behaviour of each portfolio.

7. Conclusion

Based on previous analysis, diversification helped reduce historical risk, as shown by the lower volatility, VaR and Expected Shortfall. However, comparing the return of the low-risk portfolios with the rest shows a clear trade-off between risk and return. Furthermore, the protection provided by diversification does not remain the same across all periods, varying depending on whether the events during the period affect the assets used to diversify the portfolio. The calibration performance of VaR models also depends on confidence level. The Normal GARCH and Student-t GARCH are more complex models used to estimate risk based on historical risk and how volatility responds to previous shocks, with different distributional assumptions. In particular, the latter is useful to examine shock distribution with heavier tails. However, despite considering more factors, they do not necessarily produce better calibration than the other two models, as discussed in section 5.3.

Disclaimer: Conclusions above only apply to the observed historical sample, and do not constitute investment advice.

8. References

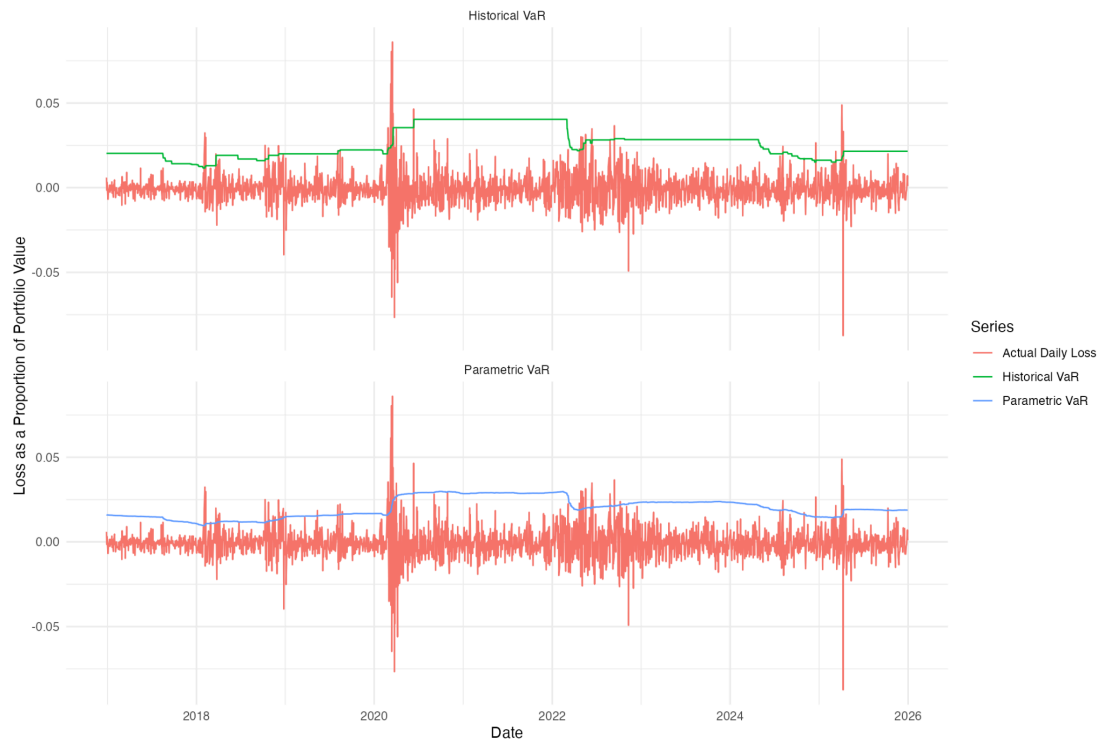
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9. Appendices

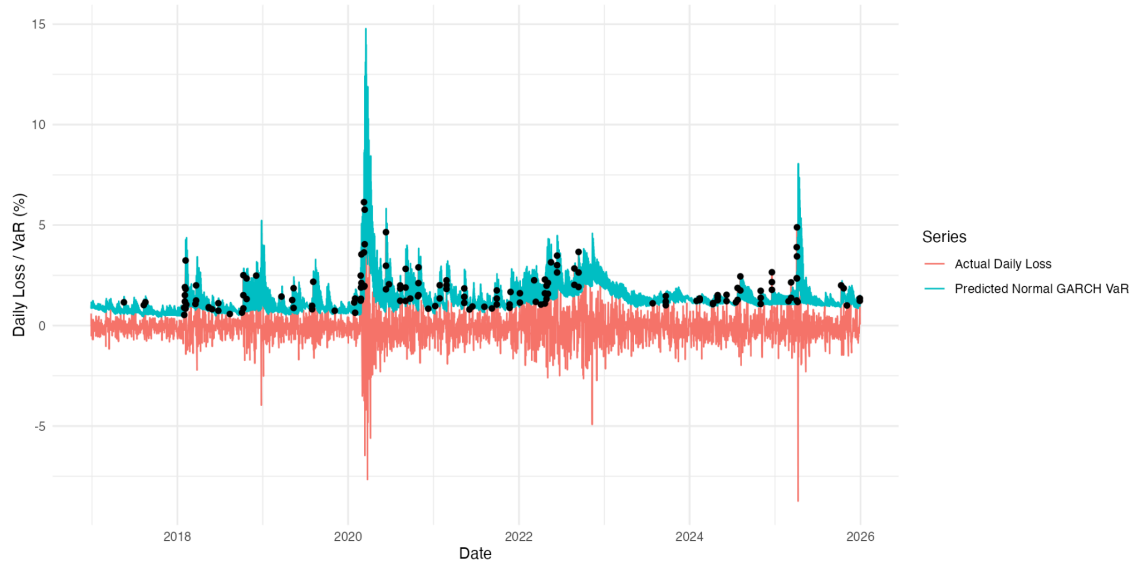
Appendix A: Additional Figures

Backtesting 99% VaR for the Equity-Heavy Portfolio
Exceedances occur when actual daily loss rises above predicted VaR

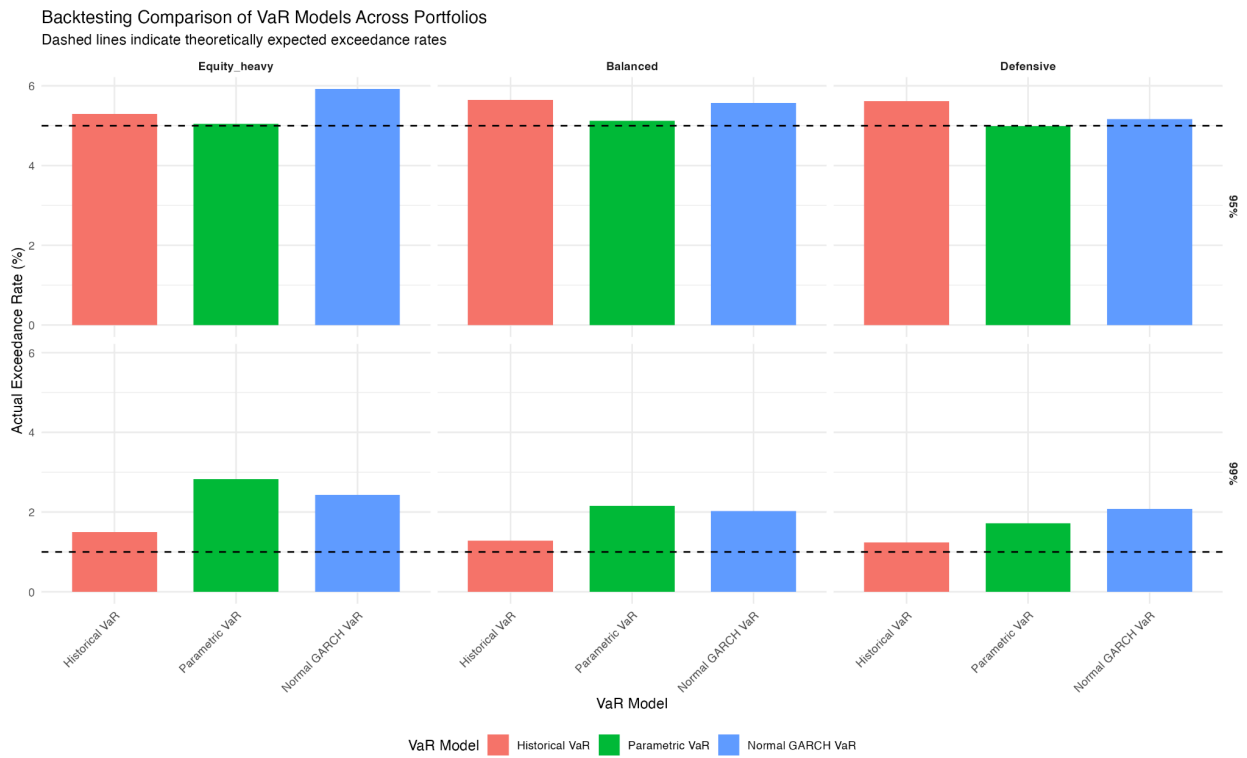


Backtesting of Historical and Parametric VaR for the equity-heavy portfolio at 99% confidence level.

Backtesting 99% Normal GARCH VaR for the Equity-Heavy Portfolio
Points indicate days when actual loss exceeded predicted VaR



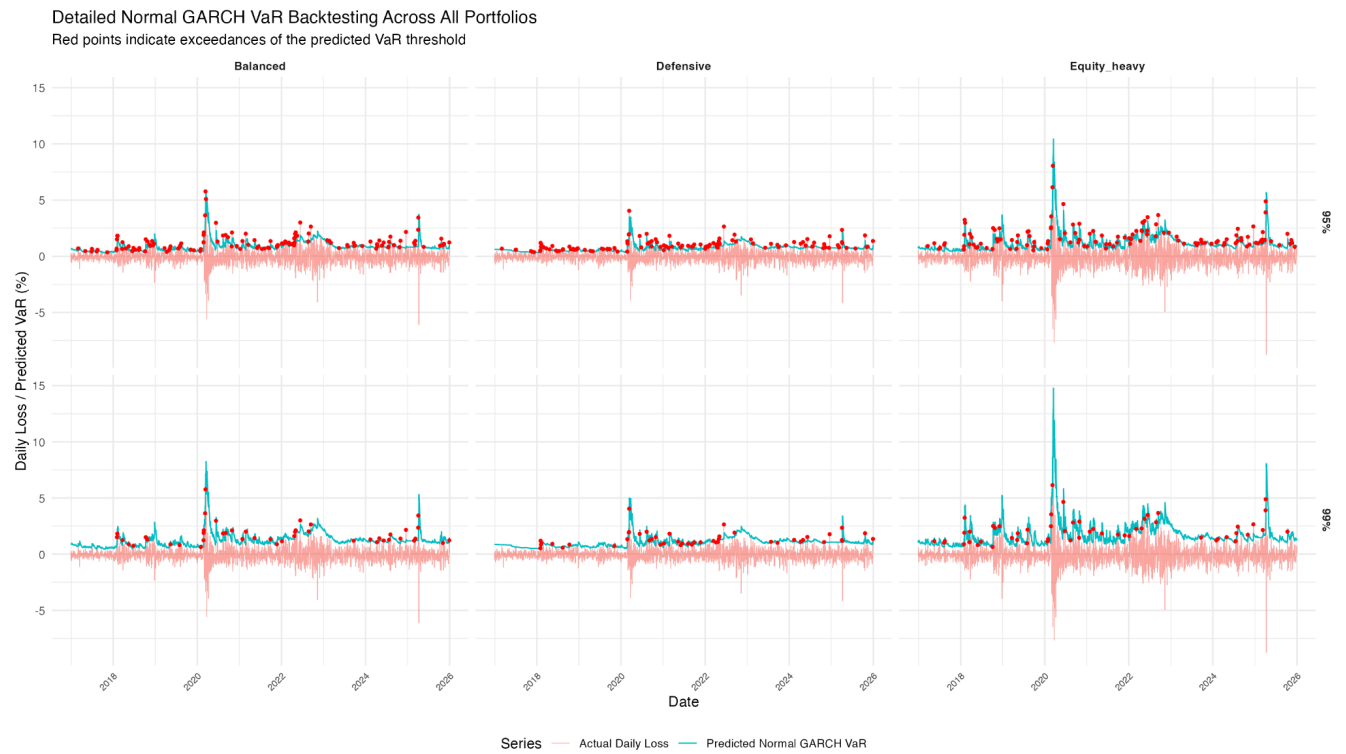
Backtesting of Normal GARCH VaR for the equity-heavy portfolio at 99%.



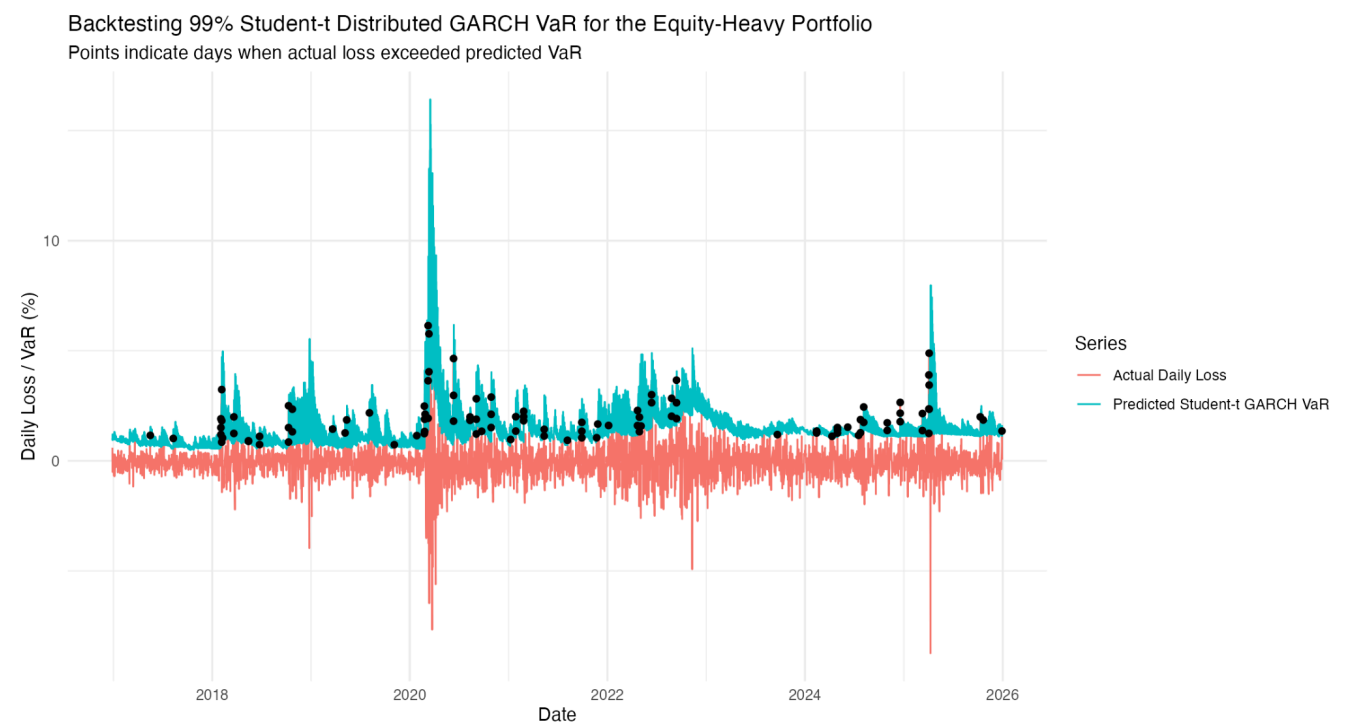
Backtesting of Historical, Parametric, and Normal GARCH VaR across portfolios and both confidence levels



Backtesting of Normal GARCH VaR across portfolios at 99% confidence level



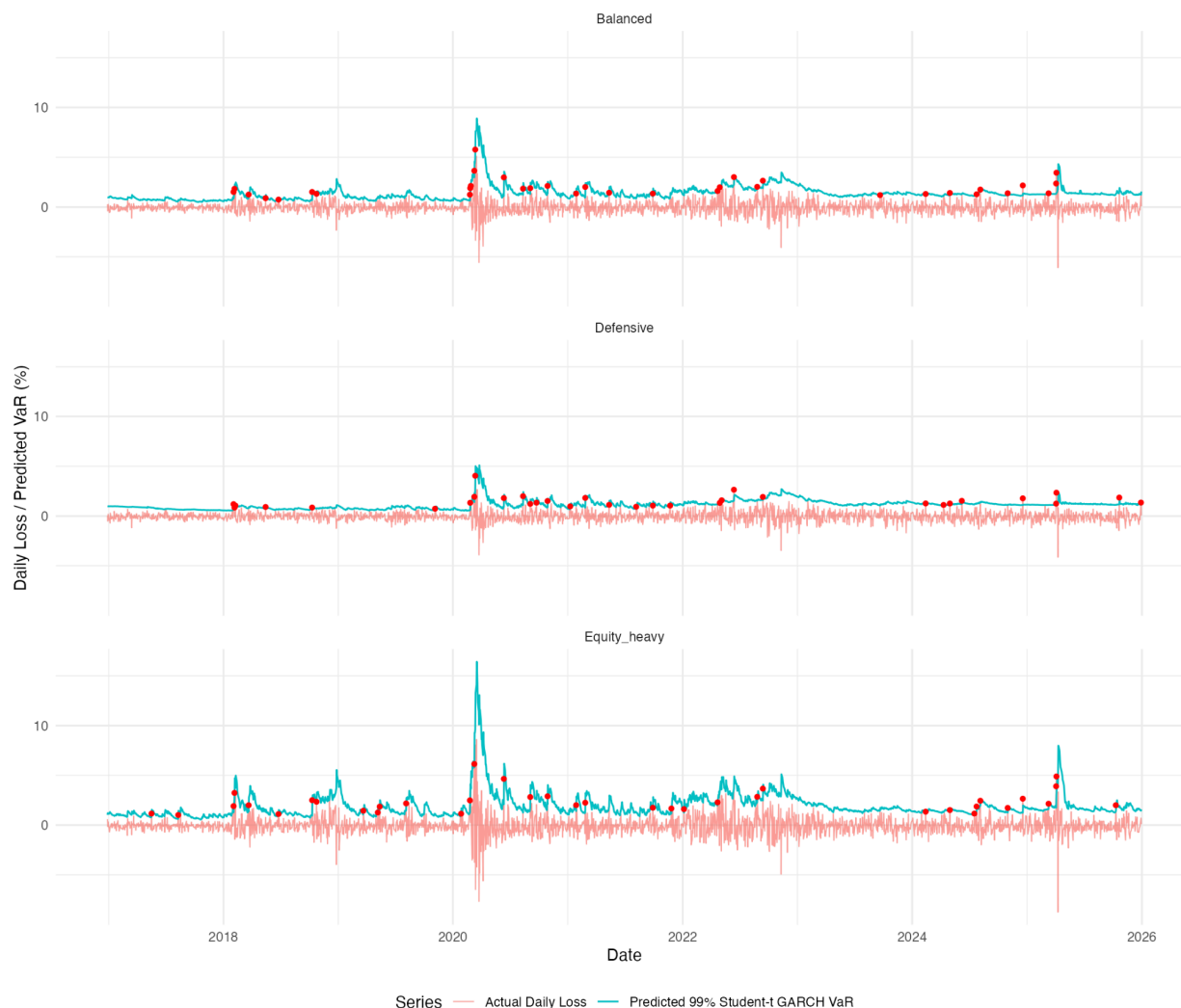
Backtesting of Normal GARCH VaR across portfolios and both confidence levels.



Backtesting of Student-t GARCH VaR for the equity-heavy portfolio at 99%

Backtesting 99% Student-t Distributed GARCH VaR Across Portfolios

Red points indicate days when actual loss exceeded predicted VaR



Backtesting of Student-t GARCH VaR across portfolios at 99%

Detailed Student-t Distributed GARCH VaR Backtesting Across All Portfolios

Red points indicate exceedances of the predicted VaR threshold



Backtesting of Student-t GARCH VaR across portfolios and both confidence levels